CIS 2450
Final Project

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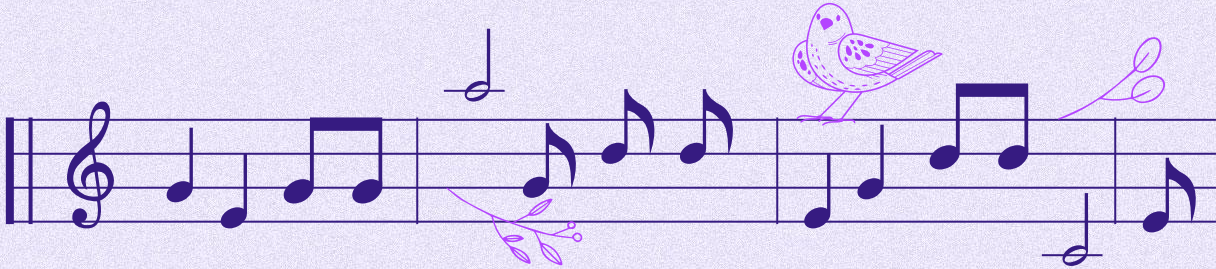
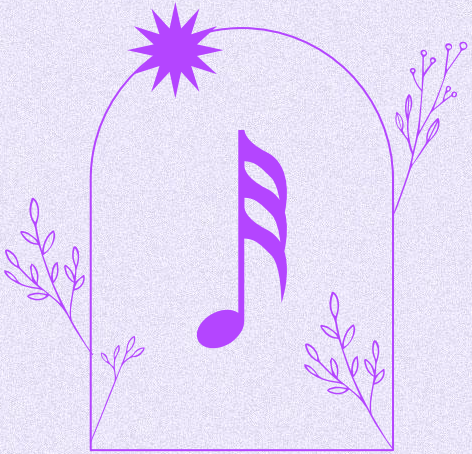


01

Project Goal

What properties of a song such as its genre, release era, duration, artist, and more will **best predict** how many times a track gets **replayed**?

Understanding Replayability



Understanding what makes listeners come back to a song over and over again is vital for producers and musicians when creating new hits.

Therefore, understanding what lyrical and acoustic factors of a song go into making a track that users play on repeat is a valuable research question.

Targets

For a given song, we wish to predict

`total_listen_count - total_user_count`

Eg, the total number of “plays” on the song minus the total number of distinct users which listen to it. This number gives us a measure of how much the song is replayed!

02

Data & EDA

Data Sources

MusicBrainz: gives us access to songs and their artists. From this source, we scrape 20 genres and over 5000 songs from each genre, giving us over 100,000 total songs to analyze.

ListenBrainz: given a song, lets us access `total_listen_count` and `total_user_count`. We can use this to measure “replayability” for the song.

AcousticBrainz: given a song, gives us access to factors like tempo (BPM), danceability, musical key, loudness, and dynamic complexity. These factors are what we will analyze to determine a song’s replayability.

Kaggle Lyrics Dataset: gives us lyrics to a song. We can use this to analyze which words are most common in high replayable songs

Preprocessing & Feature Engineering

MusicBrainz Dataset: Deduplicated on MBID so a song that surfaced under multiple searches becomes one row with its tag list collapsed into a genre_match_count feature. Engineered new temporal features (release_decade, track_age, artist_career_age) from the raw date strings, six **one-hot encoded categorical variables** (genre, release_type, artist_type, artist_country, key, key_scale) into columns while keeping the originals for display.

ListenBrainz Dataset: Joined on MBID and built the regression target repeat_listens = total_listen_count – total_user_count, then dropped rows where the join failed or the target was null.

AcousticBrainz Dataset: Coverage is only 43% (the service shut down in 2022), so we treated audio as optional enrichment instead of a hard requirement: added a has_audio_features flag and audio_feature_missing_count, and median-imputed the four numeric audio fields inside each CV fold's sklearn pipeline (no leakage from test → train). Outliers in tempo / danceability / loudness / dynamic_complexity were winsorized at the 1st and 99th percentiles, an interaction term tempo_x_dance was engineered, and all numerics were StandardScaled for the linear models.

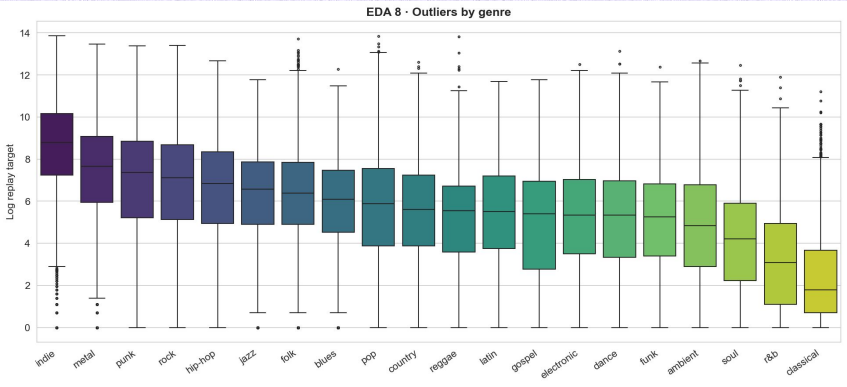
Kaggle Lyrics Dataset: Since this dataset was different from the other Brainz APIs, we had to match the lyrics from songs in the Kaggle Dataset to the MusicBrainz dataset. To do this, we used fuzzy matching based on the title and artist of the song to match them.

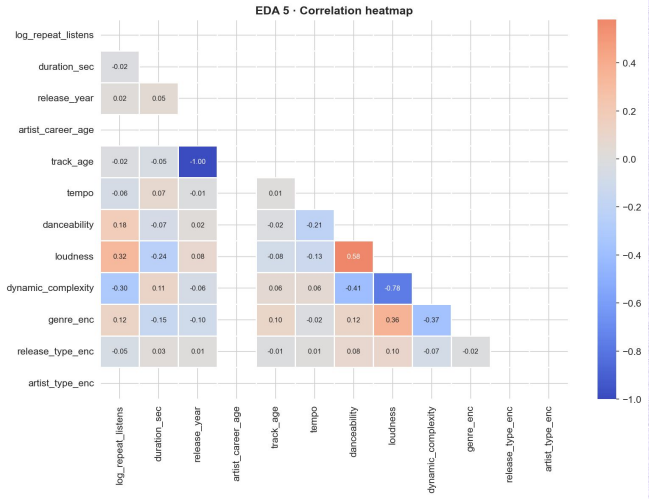
Preprocessing & Feature Engineering

We engineer the following features: **duration_sec**, **release_decade**, **track_age**, **artist_career_age**, **has_audio_features** (binary), **audio_feature_missing_count.**

To handle outliers, for the numeric features that exceed the top 1% or bottom 99% of values, we cut them off to be at the 1% outlier value so that extreme outliers don't dominate the data.

To handle correlated features, we generate a correlation matrix and remove clearly correlated features (such as **release_year** and **track_age**)





Final Data Statistics

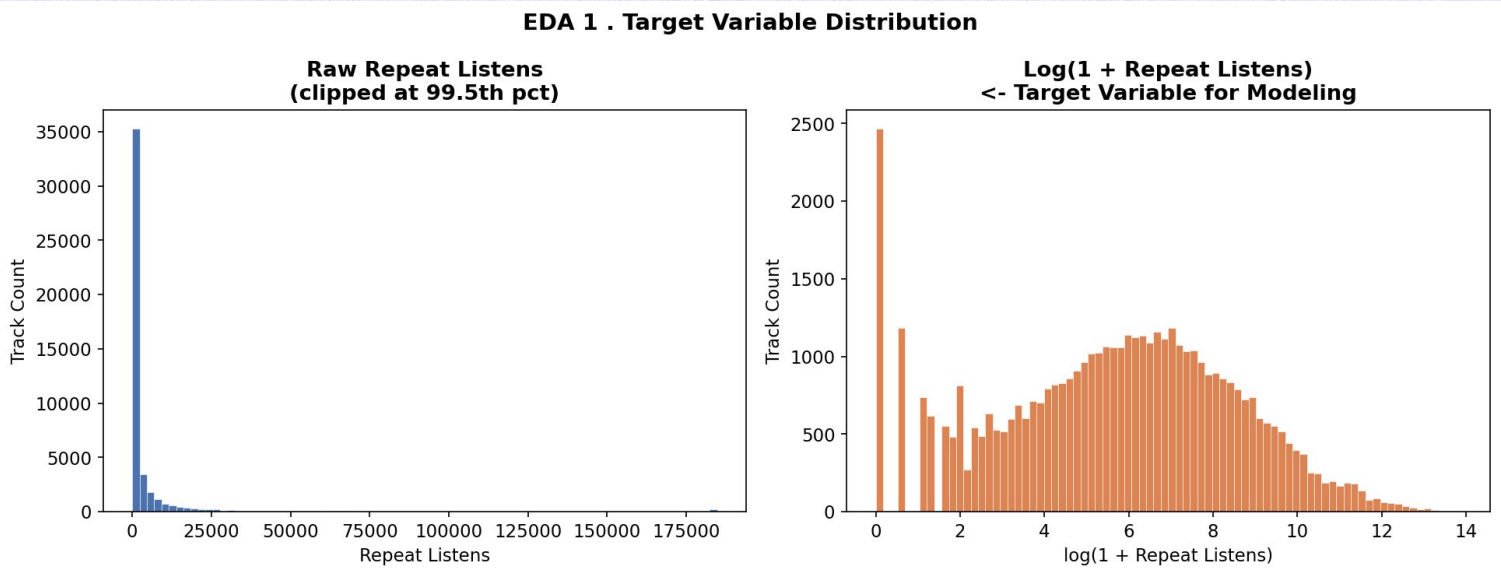
Attribute	Value
Total tracks (after cleaning)	55,579
Total columns	39
Unique artists	14,555
Genres	20
Median repeat listens	344
Median log repeat listens	5.84
AcousticBrainz coverage	42.81%
Songs matched to lyrics	20,759

EDA - MusicBrainz

MusicBrainz genres: 20

MusicBrainz songs collected: 100,000+

Raw repeat listens is very right skewed, motivating a **Log transformation**



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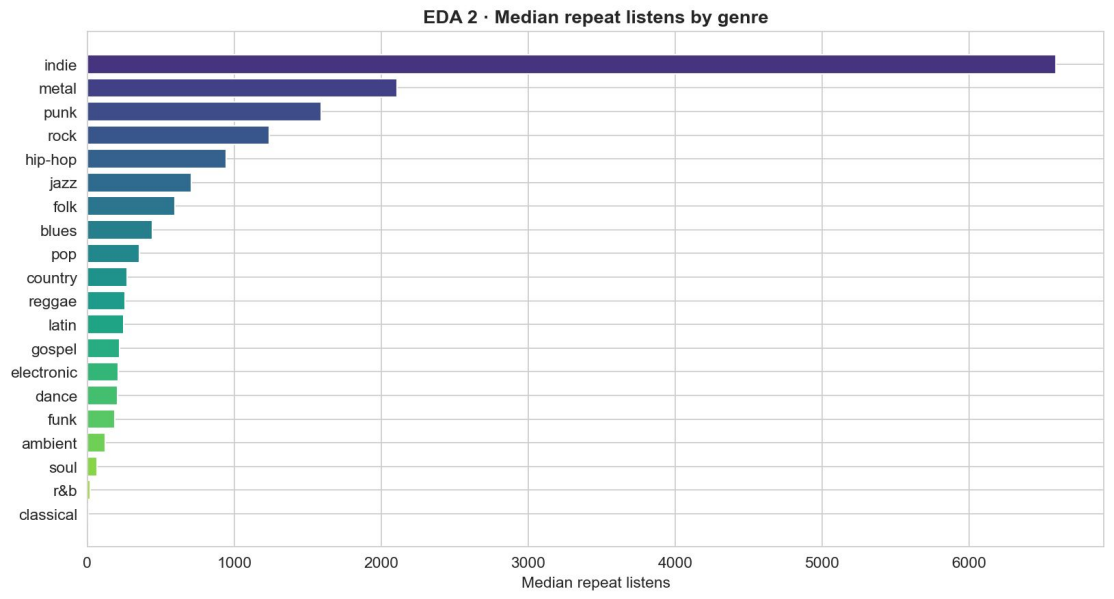
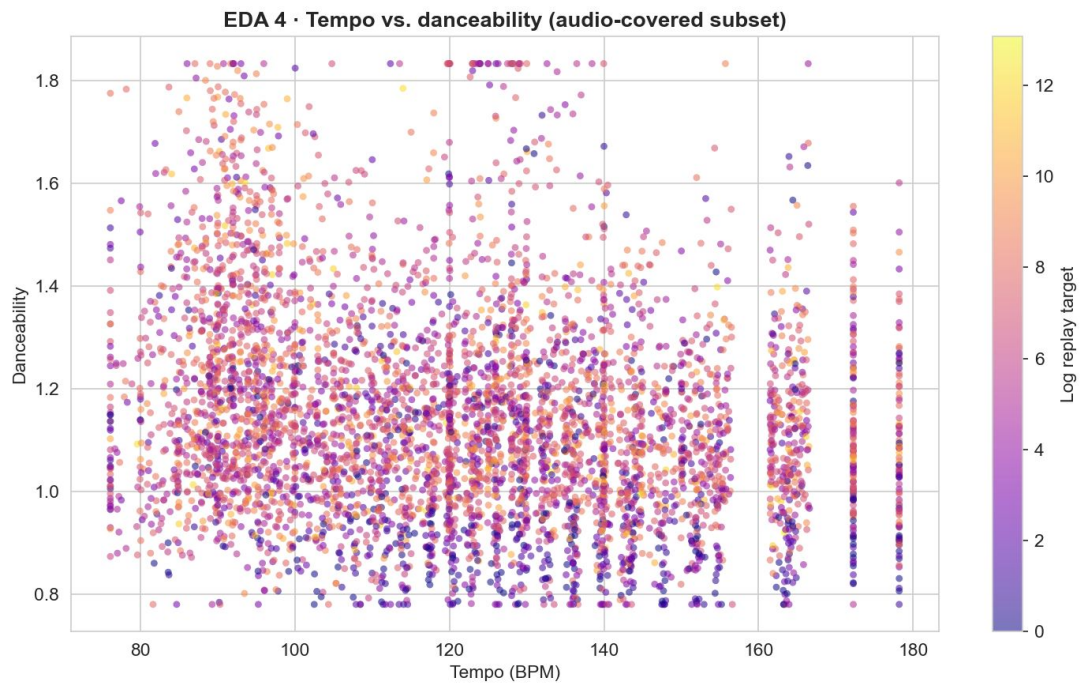
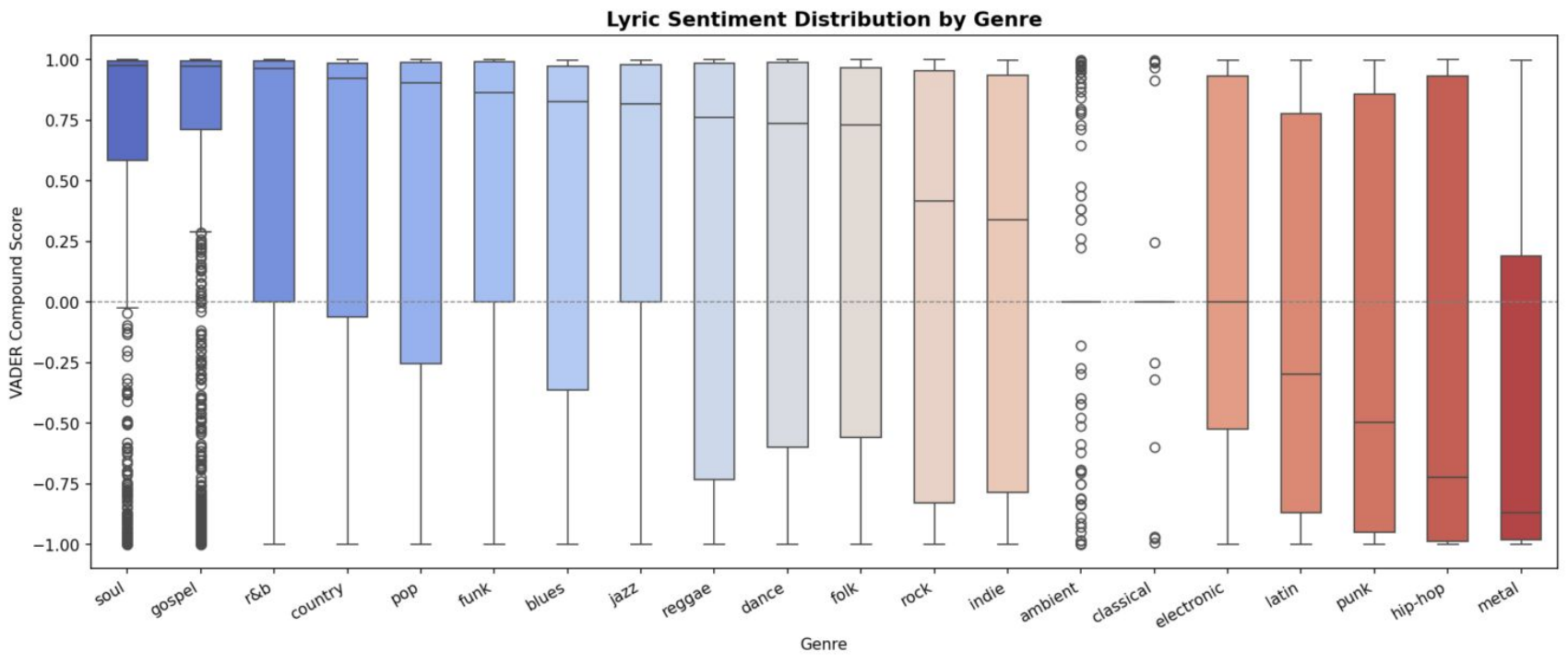


EDA-AcousticBrainz



EDA - Lyrics Data



03

Modeling

Modeling Implementation

Since our target variable is a continuous real valued number, this is a **regression problem**. Therefore, we will start by considering the following models:

Linear Regression model to establish a performance floor and test whether the relationship between features and the target variable is approximately linear.

Ridge and **Lasso** Regression to manage correlation among genre and decade features and perform implicit feature selection.

PCA + Ridge to reduce to important features and have Ridge on the components.

Gradient Boosting to capture non-linear interactions such as genre & release decade or artist type & career age that will likely influence replay behavior.

Pipeline, Model Assessment, and Hyperparameter Tuning

Model Implementation Pipeline:

Train/Test Split (80/20), 5-Fold Cross validation on training set only, Median imputation inside each fold's pipeline

Evaluation Metrics: MSE, RMSE on **Log of Replayability**, and **R^2**

We choose to evaluate on Log of Replayability since this gives us a much more normalized target variable.

Hyperparameter Tuning:

For Gradient Boosting, we used **RandomizedSearchCV** over n_estimators, learning_rate, max_depth, min_samples_leaf, subsample. This gave us better accuracy than our old untuned model.

Overall Modeling Data

Model	CV R^2	Holdout R^2	Holdout RMSE
Linear Regression (baseline)	0.247	0.232	2.505
Ridge	0.247	0.232	2.505
Lasso	0.247	0.232	2.506
PCA + Ridge	0.185	0.168	2.607
Random Forest	0.311	0.301	2.391
Gradient Boosting	0.307	0.294	2.403
Gradient Boosting (tuned)	**0.321**	**0.312**	**2.372**

Model training With Lyrics Data

We experimented with adding lyrical features to our model, such as:
sentiment_positive, sentiment_negative, avg_line_length, repetetiveness

Lyrics Prediction Model Results

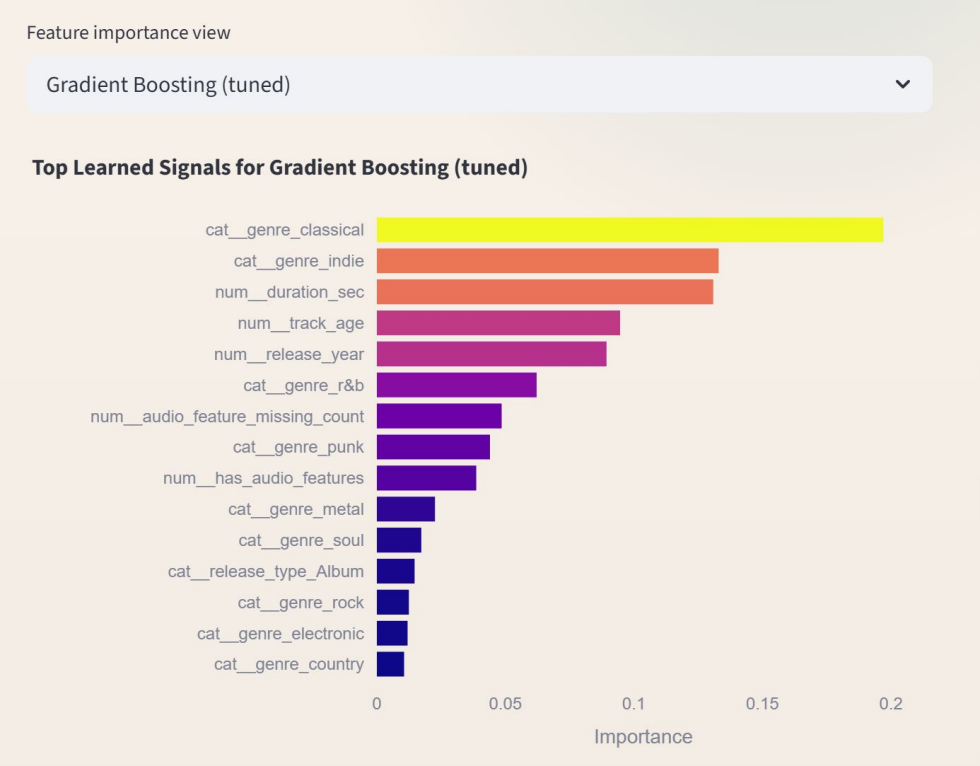
model	feature_set	algorithm	cv_r2_mean	cv_r2_std	holdout_r2	holdout_rmse	n_features	n_songs
Lyrics Only — Ridge	Lyrics Only	Ridge	0.0241	0.002	0.0334	2.502	19	23641
Lyrics Only — GBM	Lyrics Only	GBM	0.034	0.003	0.0445	2.4876	19	23641
Metadata Only — Ridge	Metadata Only	Ridge	0.0198	0.0027	0.0222	2.5165	18	23641
Metadata Only — GBM	Metadata Only	GBM	0.1927	0.0077	0.2046	2.2696	18	23641
Lyrics + Metadata — Ridge	Lyrics + Metadata	Ridge	0.0412	0.0023	0.0534	2.476	37	23641
Lyrics + Metadata — GBM	Lyrics + Metadata	GBM	0.1912	0.0077	0.208	2.2648	37	23641

Conclusion: lyrical data does not significantly affect accuracy

04

Results

Learned Feature Importance



Implications and Insights

From our feature analysis, we found:

- Genre is the single strongest predictor of replay behavior, reflecting audience loyalty patterns rather than intrinsic audio properties.
- Duration has a non-linear effect: 3–5 minute tracks replay best, consistent with the industry-standard radio-edit convention.
- Sentiment has a small but consistent negative correlation with replay ($r=-0.089$): darker lyrical themes are modestly associated with higher replays.

Course Topics Used

Polars - Data Processing Tool to aggregate MusicBrainz and Kaggle datasets
SQL - Used for EDA analytics processing
Relational Database - We store the polars dataframes in a DuckDB instance so we can query it with SQL.
Text Representations - We looked at sentiment analysis of lyrics to predict replayability.
Record Linking - We use the MBID (MusicBrainz ID) to match songs across MusicBrainz, ListenBrainz and AcousticBrainz datasets, and then fuzzy match song titles & artists to match them to the Kaggle Lyrics dataset.
Hyperparameter Tuning - RandomizedSearchCV was used to train GradientBoost model
Supervised Learning Models - All of these models were supervised learning models, since they were given a labeled training dataset
Hypothesis Testing - All of these models were trained to test different hypotheses of what factors were important in predicing replayability

Challenges & Future Work

- AcousticBrainz is a defunct service (shut down 2022)
  - 57% of tracks have no audio features, capping the model's ability to exploit tempo and danceability.

- Regression target is a proxy: ListenBrainz captures only users who actively scrobble (recording that you listen to the song on Spotify), it doesn't count for shared listens or radio listens

- Future extensions could incorporate embedding-based lyric representations.
  - More sophisticated sentence transformers

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